10.20 Renormalizability Beyond Power Counting

LecNote14 pp. 1–6 count powers of momenta, sort theories by the sign of $[g]$ (§10.14), and then make claims the counting alone does not justify. Page numbers without a LecNote refer to LecNote14. Students ask:

- "D is only 'superficial' (p. 3), and boxes on p. 4 with $D=-2$ still diverge. What does power counting guarantee?"
- "Why is the skeleton expansion of pp. 5–6 finite, and why is the last graph on p. 6 left out?"
- "Why does spin matter (p. 5)? The counting used only dimensions."
- "Infinitely many counterterms (p. 3) is 'not that terrible'?"

In short: D (here ω) is the degree when all loop momenta grow together; a subgraph can diverge on its own, and a symmetry can lower the true degree. In φ^3 at $d=6$ only tadpoles, self-energies and vertex parts are superficially divergent, so skeletons dressed with the renormalized propagator and vertex are finite. The counting assumes $1/k^2$ ($1/k$) propagators; a massive vector behaves so only when it is coupled to a conserved current. Couplings of negative dimension cost powers of k/Λ: finitely many matter at a given accuracy.

Conventions are those of §10.14 (Srednicki's, metric $(-,+,+,+)$), with $C=Z_g-1$; $\tilde{\mathbf{\Delta}}$ and V_3 are the exact φ^3 propagator and vertex; $\{\gamma^\mu,\gamma^\nu\}=-2g^{\mu\nu}$, $(\not{p}+m)u(p)=0$.

Counting with spin and derivatives

Let each field f have a propagator that falls like $k^{2[f]-d}$, as its kinetic term requires ($1/k^2$ for a scalar, $[\varphi]=\frac{d-2}{2}$; $1/k$ for a spinor, $[\Psi]=\frac{d-1}{2}$), and let a vertex with n_{if} fields f and δ_i derivatives have a coupling of dimension $[g_i]=d-\delta_i-\sum_f n_{if}[f]$. Then $\omega=dL-\sum_f(d-2[f])I_f+\sum_i\delta_i V_i$, and with $\sum_i n_{if}V_i=2I_f+E_f$, $L=\sum_f I_f-\sum_i V_i+1$,

$$\omega=d-\sum_f E_f[f]-\sum_i V_i[g_i], \tag{10.20.1}$$

which for φ^3 is (10.14.5). In $d=4$, QED ($[e]=0$) has $\omega=4-\frac{3}{2}E_\Psi-E_A$: 2, 1, 0, 0 for the vacuum polarization, the electron self-energy, the vertex and photon–photon scattering.

What ω does not say

The true degree can be lower; this is p. 3's "nice cancellations". In the one-loop QED vacuum polarization, with a Feynman parameter x,

$\ell = q - xk$ and $D = x(1-x)k^2 + m^2$, the master formula (10.18.1) (with $n = 1, 2$) makes $q^\mu q^\nu$ count as $g^{\mu\nu} D/(2-d)$; the m^2 terms cancel, and the trace $\mathrm{Tr}[\gamma^\mu(-\not{\ell} + m)\gamma^\nu(-\not{\ell} - \not{k} + m)]$ becomes $8x(1-x)(k^2 g^{\mu\nu} - k^\mu k^\nu)$:

$$\Pi^{\mu\nu}(k) = (k^2 g^{\mu\nu} - k^\mu k^\nu)\,\Pi(k^2). \tag{10.20.2}$$

Two powers of k stand outside, so $\Pi(k^2)$ is only logarithmic, and no $m^2 g^{\mu\nu}$ term calls for a photon-mass counterterm. $k_\mu \Pi^{\mu\nu} = 0$ goes with the conserved current that keeps $J \cdot A$ gauge invariant (LecNote21 p. 3); LecNote23 pp. 5–9 (Srednicki §62) derive $\Pi^{\mu\nu}$ in full.

The true divergence can also be worse. On LecNote14 p. 4 the boxes with a bubble on a side or a triangle at a corner have $\omega = -2$, but the bubble ($\omega = 2$) and the triangle ($\omega = 0$) diverge when their own loop momentum grows. The *convergence theorem* says what does hold: a Feynman integral with massive propagators, after the Wick rotation, converges if $\omega < 0$ for the whole diagram and for every subgraph. For a connected subgraph γ of a φ^3 diagram in $d = 6$, (10.14.5) gives

$$\omega(\gamma) = 6 - 2E_\gamma, \tag{10.20.3}$$

E_γ being the number of lines that attach γ to the rest. Only subgraphs attached by at most three lines are superficially divergent: tadpoles, self-energies, vertex parts. On p. 4 the counterterms already fixed by Π and V_3 cancel them: A, B on the side with the bubble, C at the corner (the graphs with $\times$). That is "exactly removed by Z_φ, Z_m, Z_g".

Skeletons (Srednicki §19; LecNote14 pp. 5–6)

A *skeleton* is a 1PI diagram with $E \geq 4$ that contains no self-energy and no vertex subgraph. Dressed, every line is $\tilde{\mathbf{\Delta}}$ and every vertex V_3, and the $2 \to 2$ amplitude is

$$T = \sum_{s,t,u} V_3^2\, \tilde{\mathbf{\Delta}} + V_4, \qquad V_4 = \sum \big(\text{dressed four-point skeletons}\big), \tag{10.20.4}$$

p. 6's first line (exchange) and second line; §10.21 evaluates it at one loop. A four-point diagram that falls apart when one line is cut is in the first line; a 1PI one, with its largest self-energy and vertex subgraphs shrunk, leaves a skeleton whose dressing contains it. When $E \geq 4$ these largest subgraphs never overlap (two that shared a line would together form a larger one, attached by at most two lines), so the skeleton is unique, and (10.20.4) expanded in α gives each ordinary diagram once (p. 6's "consistent in ordering & counting"; the table below sorts the 1PI graphs at two loops).

Each dressed skeleton is finite. Every subgraph with loops is attached by at least four lines, so by (10.20.3) it and the whole diagram have $\omega \leq -2$.

Order by order in α, dressing changes the large-momentum behaviour only by powers of logarithms: at one loop $\Pi/(k^2+m^2) \sim \frac{\alpha}{12}\ln(k^2/m^2)$ (10.14.6), and $V_3/g - 1 = -\frac{\alpha}{2}\int dF_3 \ln(D/m^2)$ (LecNote13 pp. 3–4) grows like $-\frac{\alpha}{2}\ln\lambda^2$ when all momenta scale as λ. Logarithms cannot spoil an integral with $\omega \le -2$, so V_4, and likewise every $E \ge 4$ amplitude (a tree of $\tilde{\mathbf{\Delta}}$, V_3 and skeleton sums V_n), needs no counterterm of its own.

That leaves Π and V_3, computed order by order with the lower-order counterterms inside: $\langle 0|\varphi|0\rangle = 0$ fixes Y, $\Pi(-m^2) = \Pi'(-m^2) = 0$ fix A and B, $V_3(0,0,0) = g$ fixes C (LecNote12–13). That this makes them finite at every order, even where divergent subgraphs overlap (a two-loop self-energy has two vertex subgraphs sharing a line), is the renormalization theorem, not proved here.

two-loop 1PI graph	labelled	divergent subgraph	in (10.20.4)
planar double box	6	none	two-loop skeleton
crossed double box	6	none	two-loop skeleton
box, bubble on a side	12	self-energy, $\omega = 2$	box with one-loop $\tilde{\mathbf{\Delta}}$
box, triangle at a corner	12	vertex, $\omega = 0$	box with one-loop V_3

TWO-LOOP 1PI FOUR-POINT GRAPHS OF φ^3

The table sorts LecNote14 p. 6's next order ("labelled": distinct with the legs labelled). The double boxes are new skeletons, the second line's ladders and "$+\cdots$". The other two are the dressed box with one line or one corner taken to one loop (3 orderings times 4). The graph excluded on p. 6, a box with a line across a corner, is the last row: being 1PI it is not in the first line, but it is already in the second, as the box with V_3's one-loop part at a corner.

WHY SPIN MATTERS

A massive vector, $\mathcal{L} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} - \frac{1}{2}m^2A^\mu A_\mu$, has $[A] = \frac{d-2}{2}$, but its propagator, the inverse of $(k^2+m^2)g_{\mu\nu} - k_\mu k_\nu$,

$$\tilde{\Delta}^{\mu\nu}(k) = \frac{g^{\mu\nu} + k^\mu k^\nu/m^2}{k^2 + m^2 - i\epsilon}, \tag{10.20.5}$$

tends to a constant. Each internal vector line adds 2 to (10.20.1): the vector counts as a field of dimension $d/2$, and its coupling to $\bar\Psi\gamma^\mu\Psi$ in $d = 4$ as one of dimension $4 - 3 - 2 = -1$. If the current is conserved, $k_\mu J^\mu = 0$, the $k^\mu k^\nu$ term drops out (between on-shell spinors, $\bar u(p')(\not p' - \not p)u(p) = (-m+m)\,\bar u(p')u(p) = 0$), and the counting of $[A] = 1$, $[e] = 0$ returns. Keeping the current conserved inside loops is the job of gauge symmetry: LecNote14 p. 5's "spin-1 field with gauge symmetry" (LecNote21).

Spin 2 is worse: the massive (Fierz–Pauli) propagator has a term $\frac{2}{3}k^\mu k^\nu k^\rho k^\sigma/m^4$ over k^2+m^2, so it grows like k^2. The massless one, gravity,

couples through $\kappa h_{\mu\nu}T^{\mu\nu}$, where the energy-momentum tensor has $[T^{\mu\nu}] = d$ (two derivatives, e.g. $\partial\varphi\partial\varphi$); so $[\kappa] = -\frac{d-2}{2}$, matching $[G_N] = 2 - d$ from $\frac{1}{16\pi G_N}\int d^dx\sqrt{-g}\,R$. By power counting gravity is not renormalizable (LecNote14 p. 5's red remark), but it is an effective theory, as the next subsection explains.

Not that terrible: effective theories

The superficial divergence of a diagram is a polynomial of degree ω in its external momenta and masses ($\omega + 1$ derivatives with respect to external momenta leave $\omega < 0$). Its counterterm is therefore local, with the same fields and $j \le \omega$ derivatives, and by (10.20.1) its coupling has

$$[g_{\rm ct}] = d - \sum_f E_f[f] - j \;\ge\; \sum_i V_i[g_i]. \tag{10.20.6}$$

Take couplings with $[g_i] \le 0$, those of negative dimension set by one high scale, $g_i = c_i\Lambda^{[g_i]}$ with c_i of order one. At momenta $k \ll \Lambda$ (light masses at most of order k; dimensional regularization adds no scale), (10.14.4) makes each vertex count as $g_i k^{-[g_i]} = c_i(k/\Lambda)^{-[g_i]}$. A diagram with $\sum_i V_i[g_i] = -n$ is suppressed by $(k/\Lambda)^n$, and by (10.20.6) its counterterms have $[g_{\rm ct}] \ge -n$. To accuracy $(k/\Lambda)^N$ only the finitely many terms with $[g] \ge -N$ enter, fixed by finitely many measurements: LecNote14 p. 3's effective theory, predictive for $k \ll \Lambda$ and replaced by "another theory" near Λ. Read backwards, this is why physics far below Λ looks renormalizable.

Heavy particles produce such couplings: a particle of mass M exchanged at $|k^2| \ll M^2$ gives $1/(k^2 + M^2) = M^{-2}(1 - k^2/M^2 + \cdots)$, contact terms with more and more derivatives, so $\Lambda \sim M$. Fermi's weak interaction ($[G_F] = -2$, $\Lambda \sim M_W$) has this form; gravity is the same story with $G_N \equiv 1/M_{\rm Pl}^2$ in $d = 4$. Matching at $\mu \sim M$ is in §10.18; Srednicki §29 is the general discussion.